
Education

- ✦ University of California, Santa Cruz
 - ✧ **Applied Mathematics Ph.D.** (Sept. 2024–Present), Graduate GPA: 4.0
 - ✧ **Scientific Computing & Applied Mathematics M.S.** (Aug. 2024), GPA: 4.0
 - ✧ **Mathematics B.A. with Highest Honors** (June 2023), Comp. Math., GPA: 3.74

Experience

- ✦ **Graduate Research & Teaching Assistant** — UC Santa Cruz (Sept. 2023–Present)
 - ✧ Researching rotating stratified turbulence via MPI-parallel Fortran simulations on HPC clusters; presented at APS DFD 2024 and EGU 2025
 - ✧ Teaching Assistant: AM 112 (PDEs), AM 11B, Math 19A/B; Advisor: Prof. Pascale Garaud
- ✦ **Research Assistant** — Towson University REU (Summer 2023)
 - ✧ Developed ML-based numerical methods to solve Navier-Stokes equations for ciliated organism locomotion in Stokes flow; Python implementation
 - ✧ Mentor: Dr. Hervé Nganguia
- ✦ **Peer Group Tutor** — UCSC Learning Support Services (2022–2023)

Relevant Coursework

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| ✧ Intro to Scientific ML (A) | ✧ Advanced Fluid Dynamics (A) |
| ✧ Numerical Linear Algebra (A+) | ✧ Magnetohydrodynamics (A+) |
| ✧ Numerical Methods for DEs (A+) | ✧ Computational Fluid Dynamics (A) |
| ✧ High Performance Computing (A+) | ✧ Applied & Advanced PDEs (A+) |
| ✧ Probability Theory (A-) | ✧ Stochastic Differential Equations (A) |

Technical Skills

Languages: Python, Fortran, MatLab, Java, Javascript, R

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| ✧ Scientific ML / SciML | ✧ Linear Algebra |
| ✧ Numerical Methods | ✧ Stochastic Methods |
| ✧ High Performance Computing | ✧ Technical Writing / $\text{\LaTeX}$ |

References

- + Pascale Garaud — pgaraud@soe.ucsc.edu, #:(831)459-1055
- + Hervé Nganguia — hnganguia@towson.edu, #:(410)704-2982
- + Chris Edwards — cedwards@ucsc.edu, #:(831)459-3734